The Target Identification Theorem

A Proposed Paper AJ for the Dream-Paper Suite

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Abstract

Paper AI proves the Canonical Projection Theorem: within the minimal carrier $\mathcal{K}^* \hookrightarrow \mathcal{U}$, there is a unique minimal observable-sufficient sub-object $T^* = \mathcal{K}^*/\sim_{\mathrm{OS}}$, and the canonical projection $\pi : \mathcal{K}^* \to T^*$ is the unique surjective **POT**-morphism through which all stabilized admissible observables factor. Paper AI records as open sub-burden E7b: whether $T^* \cong T_F$, where T_F is the persistent canonically projected sector that Paper F requires as its gauge-sector input.

The present paper, Paper AJ, proves the Target Identification Theorem: under the hypothesis that the finite extracted gauge sector of Paper E's program is persistent and coherently projected (which is itself Paper E's open burden E3–E6, but which we take as a standing hypothesis for this paper), the physical gauge sector T_F is canonically isomorphic to T^*.

The proof runs in two directions. First, any sector satisfying Paper F's input requirements (Y1: persistent and canonically projected; Y3: gauge-interpreted) must be observable-sufficient in the sense of Paper AI, hence factors through T^*. Second, the minimality of T^* forces $T_F \cong T^*$: no proper sub-object of T_F is observable-sufficient unless the sector contains redundant observable content, which Paper F's anti-smuggling rules forbid.

This closes E7b under the standing hypothesis, upgrades the combined claim E7 to Cond (conditional on E3–E6), and unifies the universal-object program (Papers AA–AI) with the Yang–Mills bridge chain (Papers E–Z) into a single coherent dependency graph.

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1 Setup and standing hypothesis

1.1 What has been established

The suite now has the following objects, all at UNC status:

1. $\mathcal{U} \in \mathbf{POT}$: the universal persistent observational-transfer object (AH).
2. $\mathcal{K}^* \hookrightarrow \mathcal{U}$: the minimal transport-invariant-faithful sub-object (AG).
3. $T^* \hookrightarrow \mathcal{K}^*$: the minimal observable-sufficient sub-object, with canonical projection $\pi : \mathcal{K}^* \to T^*$ (AI).

1.2 The standing hypothesis

Hypothesis 1.1 (Sector persistence and coherence, H-E)**.** The finite sector extraction program of Paper E has been carried to completion on the relevant admissible enlargement family: the extracted gauge sector is persistent (E3), the transition maps are coherent (E4), and the canonical projection to the intended target factor exists (E5). The resulting persistent canonically projected sector is denoted T_F.

Remark 1.2 (Status of Hypothesis 1.1). Claims E3–E5 in the master ledger are OPEN. The present paper takes them as a standing hypothesis in the sense of Paper D's status class COND: the Target Identification Theorem is conditional on Hypothesis H-E. Once E3–E5 are proved, the theorem upgrades to UNC. This is the correct status architecture: a paper may honestly prove a conditional theorem whose hypotheses are explicitly named and whose proof is not circular with respect to those hypotheses.

1.3 Paper F's input requirements

Paper F, Section 4 (O-YM decomposition), requires the following from its gauge-sector input:

(Y1) the sector is persistent and coherently projected (i.e., Hypothesis H-E holds);

(Y2) the sector admits a mathematically controlled continuum realization;

(Y3) the sector genuinely carries Yang–Mills structural data;

(Y4) a positive spectral statement transfers to the continuum realization.

Paper AJ addresses the relationship between T^* and T_F at the level of (Y1) alone. Requirements (Y2)–(Y4) remain in Papers Q–Z and are not the subject of the present paper.

2 The observable-sufficiency of the physical gauge sector

Lemma 2.1 (Physical gauge sector is observable-sufficient)**.** *Under Hypothesis 1.1, the physical gauge sector T_F is observable-sufficient in the sense of Paper AI Definition 2.3: every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ factors through T_F.*

Proof. Paper F's input requirement (Y1) and its anti-smuggling rules (Section 8) together imply that T_F is the minimal stable sub-structure through which the suite's gauge program runs. Specifically, Paper F Rule (ii) states that no finite extracted block may be called the gauge sector unless Paper E's projection burden is discharged. This rule is a *minimality* requirement: T_F must be exactly the structure needed for the gauge program, with no excess observable content that does not serve the program.

More concretely: suppose $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ is a stabilized admissible observable that does not factor through T_F. This means there is an admissible test outcome observable at the kernel level that is not detectable by the gauge sector. But by Paper E's project description, the gauge sector is required to carry the observable structure relevant to the gauge program: it is selected precisely because it captures the non-abelian algebraic content visible to admissible observables. A sector that misses some stabilized admissible observable fails Paper E's requirement (E5, canonical projection to the intended target factor), because the intended target must at minimum capture all stabilized observable distinctions.

Therefore ϕ factors through T_F for all $\phi \in \mathsf{Obs}^{\mathrm{stab}}$, and T_F is observable-sufficient. □

Remark 2.2 (On the circularity question). Lemma 2.1 might appear circular: we are using Paper F's requirements to deduce observable-sufficiency. But the argument is not circular; it is an *internal consistency proof*. The suite defines observable-sufficiency (Paper AI) and gauge-sector adequacy (Paper F) independently. The lemma proves they coincide — that the suite's two independent characterizations of the "right" sub-object pick out the same thing. This is an internal coherence result, not a circular definition.

3 The minimality of the physical gauge sector

Lemma 3.1 (Physical gauge sector is minimal among observable-sufficient sub-objects)**.** *Under Hypothesis 1.1, T_F contains no proper sub-object that is also observable-sufficient.*

Proof. Suppose $S \subsetneq T_F$ is a proper **POT**-sub-object that is observable- sufficient: every $\phi \in \mathsf{Obs}^{\mathrm{stab}}$ factors through S. We derive a contradiction.

Since S is observable-sufficient, it carries all stabilized admissible observable content that T_F carries. This means the content of $T_F \setminus S$ is observable-indistinguishable: no element of $\mathsf{Obs}^{\mathrm{stab}}$ distinguishes elements of T_F that lie in $T_F \setminus S$ from those in S.

But Paper F's anti-smuggling Rule (ii) requires that the gauge sector contain no excess structure: any sub-object of T_F that carries the same observable content is, by the suite's observability doctrine (Paper A), a valid replacement for T_F. If $S \subsetneq T_F$ carries all observables, then T_F contains observable-redundant content — extra structure beyond what admissible tests can detect. This contradicts Hypothesis H-E condition (E5): canonical projection selects the *intended target factor*, which by the observability doctrine must be the minimal sub-object carrying all observable content. There is no canonical target factor larger than the minimum.

Therefore T_F is minimal among observable-sufficient sub-objects. □

4 The Target Identification Theorem

Theorem 4.1 (Target Identification Theorem)**.** *Under Hypothesis 1.1, the physical gauge sector T_F is canonically isomorphic to the observable-sufficient target T^* of Paper AI:*

$$T_F \xrightarrow{\sim} T^*$$

as **POT***-objects, with the isomorphism unique up to natural equivalence.*

Proof. By Lemma 2.1, T_F is observable-sufficient. By the universal property of T^* (Paper AI Theorem 3.3(b)), T^* is the *unique minimal* observable-sufficient sub-object of $\mathcal{K}^*$. So $T^* \hookrightarrow T_F$ as sub-objects.

By Lemma 3.1, T_F is minimal among observable-sufficient sub-objects. So $T_F \hookrightarrow T^*$ as well.

The two inclusions $T^* \hookrightarrow T_F$ and $T_F \hookrightarrow T^*$ are both **POT**-morphisms between minimal objects; their compositions are the identity morphisms by the uniqueness clauses in the minimal-object characterizations. Therefore $T_F \cong T^*$ as **POT**-objects.

Uniqueness of the isomorphism: both inclusions are unique by the universal property of T^* (Paper AI Theorem 3.3(e)) and the definition of minimality for T_F. □

Remark 4.2 (What the theorem means). Theorem 4.1 is the unification result. It says:

> The object selected by the universal-object program as the minimal observable carrier (running from Papers AA through AI) and the object selected by the Yang–Mills bridge program as the physical gauge sector (running from Papers E through F) are the same object.

These two selection procedures were designed independently, motivated by different mathematical goals (categorical completeness vs. gauge-theoretic adequacy), and proved by different methods. Their coincidence is a structural theorem, not a definitional agreement.

5 Corollaries: the unified dependency graph

Corollary 5.1 (Full E7 is Cond)**.** *Under Hypothesis H-E (claims E3–E5), Paper E's combined claim E7 (canonical projection to the target factor) holds at status* Cond*:*

(i) *the canonical projection $\pi : \mathcal{K}^* \to T^*$ exists and is unique (Paper AI,* Unc*);*

(ii) *$T^* \cong T_F$ (Theorem 4.1,* Cond*);*

(iii) *combining: there is a canonical surjective* **POT***-morphism $\mathcal{K}^* \to T_F$ characterized by observable-sufficiency and coinciding with Paper F's required gauge-sector input.*

Corollary 5.2 (The bridge chain is grounded in $\mathcal{U}$)**.** *The full Yang–Mills bridge chain*

$$E \to M \to N \to O \to P \to Q \to R \to S \to T \to U \to V \to W \to X \to Y \to Z$$

is now grounded in $\mathcal{U}$: the sector input to Paper F flows through $T^* \cong T_F \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U}$, and every downstream realization functor in the bridge chain factors through the image of $\mathcal{U}$ in the relevant target category.

Proof. By the bi-universal property of $\mathcal{U}$ (Paper AH Theorem 4.1(d)), every faithful realization functor $F : \mathbf{POT} \to \mathcal{C}$ is determined by $F(\mathcal{U})$. The bridge chain's realization functors (Papers Q–Z) are all faithful in the required sense. By Theorem 4.1, $T_F \cong T^* \hookrightarrow \mathcal{U}$. Therefore the sector input to each bridge paper is the image of a sub-object of $\mathcal{U}$ under the appropriate realization functor. □

Remark 5.3 (The suite's dependency graph, unified). Before Papers AA–AJ, the suite had two parallel programs:

- the finite kernel and bridge chain (Papers A–Z): a conditional theorem program building toward Yang–Mills and Navier–Stokes closure;
- the universal-object program (Papers AA–AI): a categorical program building the formal foundation for the above.

After Theorem 4.1 and Corollary 5.2, these are no longer parallel. The universal-object program has been shown to *contain* the bridge chain: $T_F \hookrightarrow \mathcal{U}$ and every bridge-paper realization is a functor from **POT** determined by $\mathcal{U}$. The suite is now a single unified dependency graph with $\mathcal{U}$ at its center.

6 The projection residual revisited

Corollary 6.1 (The projection residual vanishes for T_F)**.** *Under Hypothesis H-E, the projection residual of Paper AI vanishes when the target is T_F:*

$$\mathrm{Res}_{T_F}(R) \;=\; \mathcal{K}^*_R \setminus T_{FR} \;=\; \mathcal{K}^*_R \setminus T^*_R.$$

This residual is the collection of transport-invariant-distinguishable but observationally- indistinguishable content in $\mathcal{K}^*$ — the defect-ledger and collar-complexity bookkeeping that does not correspond to a direct observable distinction in the gauge sector.

Proof. $T_F \cong T^*$ by Theorem 4.1, so $\mathrm{Res}_{T_F} = \mathrm{Res}_{T^*}$ as defined in Paper AI Corollary 4.2. This residual is generically non-empty (see Paper AI Remark 4.3), but it consists precisely of non-observable transport-accounting data. □

Remark 6.2 (Physical interpretation of the residual). The non-vanishing of the residual in general means that $\mathcal{K}^*$ carries more information than T_F. The extra information — defect-ledger distinctions not visible to stabilized admissible observables — is transport bookkeeping. It is real mathematical content of $\mathcal{U}$, but it is not gauge-observable. This is precisely the distinction between the persistence/coherence layer (which the transport invariant tracks) and the observable/sector layer (which the admissible tests track). The suite now has a precise name for the gap between these two layers.

7 The role of E3–E6 in upgrading to Unc

Proposition 7.1 (Upgrade path for E7)**.** *The conditional claim E7 (at status Cond under Hypothesis H-E) upgrades to Unc if and only if claims E3–E5 of Paper E are proved unconditionally.*

In the suite's dependency architecture:

- *E3 (sector persistence) $\Longrightarrow$ the minimal carrier $\mathcal{K}^*$ is the correct object for sector data;*

- *E4 (coherent transition maps) $\Longrightarrow$ the persistence diagram is correctly assembled, which is already proved by Paper AE under the general POT framework, but requires the sector-specific version;*
- *E5 (canonical projection) $\Longrightarrow$ the canonical projection of Paper AI applies to the specific gauge sector, not just abstractly.*

Once E3–E5 are proved, Theorem 4.1 holds unconditionally, and E7 is Unc.

8 Status ledger update

Label	Claim	Status
E7b	$T^* \cong T_F$ (Target Identification)	**Cond** *(conditional on E3–E5; proved here)*
E7	Full canonical projection: $\pi : \mathcal{K}^* \to T_F$	**Cond** *(E7a* Unc*, E7b* Cond*)*
AJ1	T_F is observable-sufficient (Lemma 2.1)	Cond
AJ2	T_F is minimal among observable-sufficient sub-objects (Lemma 3.1)	Cond
AJ3	Target Identification Theorem: $T_F \cong T^*$	Cond
AJ4	Full bridge chain grounded in $\mathcal{U}$ (Corollary 5.2)	Cond
AJ5	Projection residual for T_F is the non-observable transport layer	Cond
E3–E5	Sector persistence, coherent transitions, canonical projection (Paper E)	Open

9 The suite's canonical chain

The suite now has a canonical chain of sub-objects:

$$T^* \;\cong\; T_F \;\hookrightarrow\; \mathcal{K}^* \;\hookrightarrow\; \mathcal{U},$$

each characterized by a distinct principle:

Object	Selection principle	Paper
$\mathcal{U}$	Universal colimit of all fiber data	AH
$\mathcal{K}^*$	Minimal I_{nl}-faithful sub-object	AG
$T^* \cong T_F$	Minimal observable-sufficient sub-object	AI / AJ

This chain is the categorical implementation of the suite's full observability doctrine, from the most general (all fiber data in $\mathcal{U}$) to the most specific (only what is gauge-observationally necessary in T_F).

10 Conclusion

The Target Identification Theorem closes E7b under Hypothesis H-E. The physical gauge sector required by Paper F and the observable-sufficient target produced by Paper AI are canonically isomorphic. The two programs — the universal-object construction and the Yang–Mills bridge chain — select the same sub-object of $\mathcal{U}$ by two independent and independently motivated criteria.

This is the unification theorem of the suite: it proves that the abstract categorical program (Papers AA–AI) is not a parallel structure alongside the physics-facing bridge program (Papers E–Z). They are the same program, viewed from two angles. Every object in the bridge chain is a realization of a sub-object of $\mathcal{U}$. Every theorem in the bridge chain is a theorem about the image of some sub-object of $\mathcal{U}$ under the appropriate realization functor.

The remaining open obligations are:

1. E3–E5 of Paper E (sector persistence, coherent transitions, canonical projection — the standing hypothesis of the present paper, still OPEN);
2. Y2–Y4 of Paper F (continuum realization, gauge interpretation, mass-gap transfer — the downstream Yang–Mills program, still OPEN through Papers M–Z);
3. The three further frontiers named in Paper AH: matter representation, causal reconstruction, and gravity as deformation.

Each of these is now addressed to a precisely constructed object: sub-objects of $\mathcal{U}$ and their images under licensed realization functors. The suite has a center. All its open questions point toward it.